PCSE – REAL LIFE PROJECT 1 & 2

- Expectations?
- Professional Certificate in Software Engineering
- Industry-Oriented Practical Assessment Guide
- For Academic Delivery & Student Reference

UNDERSTANDING COMPETENCE-BASED EDUCATION (CBE)

- Focus on what students can DO
- Practical assessment over theory-heavy exams
- Industry relevance and employability focus

WHY REAL-LIFE PROJECTS EXIST IN PCSE

- To simulate real software development environments
- To test independent thinking
- To build confidence and innovation
- To prepare students for employment or entrepreneurship

REAL LIFE PROJECT 1 – CORE OBJECTIVE

- Demonstrate mastery of foundational programming
- Identify and solve a simple real-world problem
- Develop a complete working system
- Submit structured documentation

SKILLS DEVELOPED IN PROJECT 1

- Problem analysis
- Algorithm development
- Basic UI design
- Testing fundamentals
- Report writing skills

REAL LIFE PROJECT 1 – EXPECTED DELIVERABLES

- Project proposal
- System design diagrams
- Working software application
- Testing evidence
- Printed project report

REAL LIFE PROJECT 2 – CORE OBJECTIVE

- Demonstrate advanced system development skills
- Integrate multiple technologies
- Apply software engineering lifecycle
- Work with minimal supervision

SKILLS DEVELOPED IN PROJECT 2

- Database design and integration
- User authentication and validation
- Modular architecture
- Error handling & debugging
- Professional documentation

SOFTWARE DEVELOPMENT LIFECYCLE (SDLC) APPLICATION

- Problem identification
- Requirements analysis
- System design
- Implementation
- Testing
- Deployment
- Maintenance awareness

COMPARING PROJECT 1 AND PROJECT 2

- Project 1 – Guided & Basic
- Project 2 – Independent & Advanced
- Project 2 must show higher complexity
- Project 2 demands stronger technical justification

PROJECT SELECTION GUIDELINES

- Choose real community problems
- Avoid copied internet systems
- Ensure project scope is manageable
- Consult lecturer before approval

DOCUMENTATION STRUCTURE (CHAPTER FORMAT)

- Chapter 1 – Introduction
- Chapter 2 – Literature Review
- Chapter 3 – Methodology
- Chapter 4 – System Implementation
- Chapter 5 – Summary & Recommendations

SYSTEM DESIGN REQUIREMENTS

- Flowcharts
- Use Case Diagrams
- ER Diagrams (Project 2)
- Database schema design

CODING STANDARDS EXPECTATIONS

- Meaningful variable names
- Code comments
- Modular functions
- Readable formatting

TESTING REQUIREMENTS

- Unit testing
- Functional testing
- Error handling testing
- User acceptance testing

SECURITY CONSIDERATIONS (PROJECT 2)

- Password hashing
- Input validation
- Data protection awareness

ASSESSMENT RUBRIC (EXPANDED)

- Problem relevance – 15%
- System design – 20%
- Implementation quality – 30%
- Documentation – 15%
- Testing – 10%
- Defense & Presentation – 10%

GRADING DESCRIPTORS

- Distinction – Excellent functionality & documentation
- Credit – Good system with minor weaknesses
- Pass – Basic functionality achieved
- Fail – Incomplete or non-functional system

COMMON CAUSES OF FAILURE

- Plagiarism
- Incomplete implementation
- Weak documentation
- Poor presentation skills

PROJECT DEFENSE EXPECTATIONS

- Explain system architecture
- Justify technology choice
- Demonstrate live system functionality
- Answer technical questions confidently

SAMPLE VIVA (PROJECT DEFENSE) QUESTIONS

- Why did you choose this problem?
- Explain your database structure.
- What challenges did you face?
- How would you improve the system?

PROFESSIONAL CONDUCT DURING PRESENTATION

- Clear communication
- Time management (10–15 minutes)
- Respect panel feedback

PORTFOLIO DEVELOPMENT

- Host project on Site Provided (if applicable)
- Prepare demo screenshots
- Add project to your Profile/CV
- Prepare project summary for interviews

CAREER RELEVANCE OF REAL-LIFE PROJECTS

- Internship readiness
- Freelancing opportunities
- Startup foundation
- Industry job preparedness

FINAL ACADEMIC ADVICE

- Start early
- Be eager to learn at all times
- Work hard
- Work consistently
- Avoid shortcuts
- Aim for excellence, not minimum pass

CONCLUSION

- Real Life Projects define your professional identity.
- Treat them as career investments.

EXAMPLE PROJECTS – REAL LIFE PROJECT 1

1. Student Result Processing System
   - Input marks and calculate grades
   - Generate printable report
   - Basic database or file storage
   - Skills: Logic, validation, simple UI

PROJECT 1 EXAMPLE – INVENTORY SYSTEM

2. Inventory Management System (Small Shop)
   - Record products and stock levels
   - Alert when stock is low
   - Generate sales summary
   - Skills: CRUD operations, basic database usage

PROJECT 1 EXAMPLE – LIBRARY SYSTEM

3. Library Management System
   - Register books and members
   - Issue and return books
   - Fine calculation
   - Skills: OOP basics, file/database handling

EXAMPLE PROJECTS – REAL LIFE PROJECT 2

1. Online E-Commerce System
   - User authentication and login
   - Shopping cart and order processing
   - Admin dashboard
   - Skills: Authentication, session management, MVC structure

PROJECT 2 EXAMPLE – HOSPITAL MANAGEMENT

2. Hospital Management System
   - Patient records and scheduling
   - Billing and pharmacy module
   - Role-based access control
   - Skills: ER design, multi-user system, security validation

PROJECT 2 EXAMPLE – SACCO SYSTEM

3. SACCO / Microfinance Management System

- Member registration
- Loan processing and savings tracking
- Financial reporting
- Skills: Complex calculations, secure database transactions

PROJECT COMPLEXITY COMPARISON

- Project 1: Basic, guided, small scope
- Project 2: Advanced, independent, integrated modules
- Project 2 May require authentication, normalized database, modular design

COMMUNITY-BASED PROJECT IDEAS

- Market Vendor Sales System
- School Attendance System
- Youth SACCO Management System
- Rental Property Management System
- Etc.
- You are encouraged to solve local community problems

YOUR PROJECT = YOUR PROFESSIONAL PORTFOLIO

- Portfolio-ready
- Interview-ready
- Demo-ready
- Deployment-ready
- Treat your Real-Life Project as your first software company.